

Muhammad Ibrahim

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EDUCATION

Union College

Bachelor of Science in Computer Science & Mathematics (Double Major)

Schenectady, NY

Sep. 2024 – Jun. 2028

Relevant Coursework: Data Structures, Abstract Algebra, Computational Game Theory, Calculus, Geometry, Logic and Set Theory

EXPERIENCE

Templeton Institute – Sexagesimal Research

Research Assistant

Schenectady, NY

Sep. 2024 – Present

- Conducting funded research transcribing Persian and Arabic mathematical manuscripts into base-60 computation
- Implemented 6+ arithmetic operations (addition, subtraction, multiplication, division, modulo, exponentiation) in a Python iterator engine
- Engineered a Python-based sexagesimal expression parser, bridging ancient numeral systems with modern computational logic

iTech Solutions

Information Technology Intern

Pakistan

Aug. 2023 – Present

- Assisted a 3–4 member IT team with routine system checks, troubleshooting, and technical support for 10+ office staff
- Entered, updated, and verified ~200 data records across departments to maintain accurate internal documentation

PROJECTS

Stock Price Predictor | Python, scikit-learn, Pandas, yfinance

2025 – 2026

- Built an ML pipeline to forecast stock closing prices using Linear Regression and Random Forest across 5+ years of historical data, achieving 87% directional accuracy on held-out test sets
- Engineered 10+ time-series features including rolling averages and volatility indicators, reducing RMSE by 23% over the baseline model

PathFinder | TypeScript, React, Flask, OpenAI API

2025 – 2026

- Built a full-stack course planning web app with a TypeScript/React frontend and Flask REST API backend, serving GPT-powered academic recommendations across 100+ Union College courses
- Leveraged TypeScript interfaces and strict typing to reduce runtime errors by 40%, enabling real-time major requirement tracking and schedule conflict detection for 50+ beta users

CampusQuery | SQL, Python, SQLite

2025

- Built a normalized relational database with 3+ interconnected tables, writing SQL queries using joins, aggregations, and filters to extract structured academic insights
- Automated query pipelines via Python across 200+ records using normalization and referential integrity principles

Poker Game Simulation | Java, JUnit

2025

- Engineered a full poker game engine implementing community cards, player actions, and core game logic within a collaborative team
- Validated correctness across edge cases using assertion-based testing and systematic debugging

ACTIVITIES

Union College Cricket Club – Captain

Aug. 2025 – Present

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript, HTML/CSS

Frameworks & Libraries: React, Flask, scikit-learn, Pandas, Matplotlib, NumPy

Developer Tools: Git, SQLite, LaTeX, Microsoft Excel, VS Code

Concepts: Object-Oriented Programming, Machine Learning, Data Structures, Database Design, REST APIs